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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 9(a) | <p>any <b>seven</b> from</p> <ol style="list-style-type: none"> <li>1 (H<sup>+</sup>) carrier / pump (protein), in cell surface membrane (of guard cell) ;</li> <li>2 hydrogen ions / protons / H ions / H<sup>+</sup>, leave / exit (cell) ;</li> <li>3 using, energy / ATP ;</li> <li>4 low H<sup>+</sup> (in cell) / more negative charge (than outside) ;</li> <li>5 K<sup>+</sup> channel (proteins) open ;</li> <li>6 K<sup>+</sup>, move into / enter, cell (by facilitated diffusion) ;</li> <li>7 Cl<sup>-</sup> ions, move into / enter (cell) ;</li> <li>8 water / solute, potential of cell decreases ;</li> <li>9 water moves in (to cell) by <u>osmosis</u> ;</li> <li>10 cell / vacuole, <u>volume</u> increases ;</li> <li>11 (cells) are / become, turgid ;</li> <li>12 thick inner cell wall (of guard cell) ;</li> <li>13 AVP ;</li> </ol> | 7 |
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| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
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| 9(b)     | <p>any <b>eight</b> from:</p> <ol style="list-style-type: none"> <li>1 C4, plants / pathway ;</li> <li>2 stop / decrease, <u>photorespiration</u> ;</li> <li>3 mesophyll cells form (tightly packed) ring ;</li> <li>4 around / surrounding, bundle sheath (cells) ;</li> <li>5 (initial) CO<sub>2</sub>/carbon, fixation in mesophyll and Calvin cycle in bundle sheath ;</li> <li>6 RuBP / rubisco, in bundle sheath (cells) ;</li> <li>7 oxygen / air, cannot reach, RuBP / rubisco / bundle sheath ;</li> <li>8 CO<sub>2</sub> combines with, PEP / PEP carboxylase (in mesophyll) ;</li> <li>9 to form, oxaloacetate / malate ;</li> <li>10 (malate) releases carbon dioxide in bundle sheath (cells) ;</li> <li>11 RuBP, carboxylated / reacts with carbon dioxide ;</li> <li>12 PEP carboxylase has high optimum temperature / tolerates high temperatures ;</li> </ol> | 8     |